A Metric-Approximation Subcase of Semiprimeness for Free Banach f-Algebras

Agent lane 11

2026-07-19

Status

This is a partial result for Question 5.10 in David Munoz-Lahoz and Pedro Tradacete's paper *Free Banach f-algebras* [1]. No full solution of the source question is claimed.

Source Metadata

- Source arXiv id: 2511.13299.
- Source title: *Free Banach f-algebras*.
- Source authors: David Munoz-Lahoz and Pedro Tradacete.
- Source question location: Question 5.10, PDF page 66.

Open Problem Evidence

(1) *the closure of* $\bigcup_\lambda E_\lambda$ *is* E,
(2) *there exist contractive projections* $P_\lambda \colon E \to E_\lambda$,
(3) *the free norm in* $\mathrm{FB}f\mathrm{A}[E_\lambda]$ *coincides with* τ_{E_λ}.

Then $\mathrm{FB}f\mathrm{A}[F]$ *is representable in* $C(B_{F^*})$.

Proof. By Proposition 4.32, τ is a norm on the Banach f-algebras generated by these Banach spaces. The result follows from a successive application of Proposition 5.6 and Theorem 5.1. □

Corollary 5.8. $\mathrm{FB}f\mathrm{A}[L_1(\mu)]$ *is representable in* $C(B_{L_\infty(\mu)})$ *for every measure* μ.

Remark 5.9. In particular, everything we have done so far applies to $\mathrm{FB}f\mathrm{A}(S) = \mathrm{FB}f\mathrm{A}[\ell_1(S)]$ for every set S. Hence, from the point of view of the free Banach f-algebra generated by a set, we are being completely general.

After this discussion, we have to leave the following open.

Question 5.10. Is $\mathrm{FB}f\mathrm{A}[E]$ semiprime for every Banach space E?

The source question is:

$$\text{Is } \mathrm{FB}f\mathrm{A}[E] \text{ semiprime for every Banach space } E?$$

The source proves positive answers for finite-dimensional E and for $E = L_1(\mu)$, and proves that semiprimeness of $\mathrm{FB}f\mathrm{A}[E]$ is equivalent to representability in $C(B_{E^*})$ [1, Theorem 5.1 and Corollaries 5.2, 5.8].

Partial Result

Say that E has the metric approximation property, in the net form used here, if there is a directed net (R_λ) of finite-rank operators $R_\lambda : E \to E$ such that

$$\|R_\lambda\| \le 1 \quad \text{and} \quad R_\lambda x \longrightarrow x \quad \text{for every } x \in E.$$

Proof Intuition

The finite-dimensional pieces already have semiprime free Banach f-algebras by the source paper. The approximating maps do not have to be projections. A finite-rank contraction $R_\lambda : E \to E$ factors as a contraction from E onto its finite-dimensional range, followed by the inclusion of that range back into E. If $a \in \mathrm{FB}f\mathrm{A}[E]$ satisfies $a^2 = 0$, the first factor sends a to a nilpotent element in a finite-dimensional free Banach f-algebra, so it kills a. Therefore the induced endomorphism associated to R_λ kills a. Since these endomorphisms converge strongly to the identity on $\mathrm{FB}f\mathrm{A}[E]$, the element a is a norm limit of zero elements.

Theorem 1. *Let E be a Banach space with the metric approximation property. Then $\mathrm{FB}f\mathrm{A}[E]$ is semiprime. Consequently, the canonical representation of $\mathrm{FB}f\mathrm{A}[E]$ in $C(B_{E^*})$ is injective.*

Proof

We first record the functorial continuity point used in the argument.

Lemma 1. *Let $T_\lambda : E \to E$ be contractions such that $T_\lambda x \to x$ for every $x \in E$. Let $\overline{T_\lambda} : \mathrm{FB}f\mathrm{A}[E] \to \mathrm{FB}f\mathrm{A}[E]$ be the contractive lattice-algebra homomorphism induced by the universal property. Then*

$$\overline{T_\lambda} a \longrightarrow a \quad \textit{for every } a \in \mathrm{FB}f\mathrm{A}[E].$$

Proof. The normed free algebra $\mathrm{FN}f\mathrm{A}[E]$ is dense in $\mathrm{FB}f\mathrm{A}[E]$. Every element $b \in \mathrm{FN}f\mathrm{A}[E]$ is obtained from finitely many generators $\eta_E(x_1), \dots, \eta_E(x_n)$ by finitely many linear, lattice, and algebra operations. Since $T_\lambda x_j \to x_j$ for each j, and since the Banach lattice and Banach algebra operations are norm continuous, $\overline{T_\lambda} b \to b$.

For general $a \in \mathrm{FB}f\mathrm{A}[E]$, choose $b \in \mathrm{FN}f\mathrm{A}[E]$ with $\|a-b\| < \varepsilon$. The maps $\overline{T_\lambda}$ are contractions, so

$$\|\overline{T_\lambda} a - a\| \le \|\overline{T_\lambda}(a-b)\| + \|\overline{T_\lambda} b - b\| + \|b-a\| \le 2\varepsilon + \|\overline{T_\lambda} b - b\|.$$

Letting λ tend along the net and then $\varepsilon \downarrow 0$ proves the claim. □

Now assume $a \in \mathrm{FB}f\mathrm{A}[E]$ satisfies $a^2 = 0$. Let (R_λ) be a net of finite-rank contractions on E converging strongly to the identity. For each λ, put $E_\lambda = R_\lambda(E)$, equipped with the norm inherited from E, and view R_λ as a contraction

$$S_\lambda : E \longrightarrow E_\lambda.$$

By the universal property, S_λ induces a contractive lattice-algebra homomorphism

$$\widetilde{S_\lambda} : \mathrm{FB}f\mathrm{A}[E] \longrightarrow \mathrm{FB}f\mathrm{A}[E_\lambda].$$

Since this is an algebra homomorphism,

$$(\widetilde{S_\lambda} a)^2 = \widetilde{S_\lambda}(a^2) = 0.$$

The source paper proves that $\mathrm{FB}f\mathrm{A}[F]$ is semiprime whenever F is finite-dimensional [1, Corollary 4.23]. Hence $\widetilde{S_\lambda} a = 0$ for every λ.

Let $i_\lambda : E_\lambda \hookrightarrow E$ be the inclusion. The composition $i_\lambda S_\lambda : E \to E$ is exactly R_λ, now regarded as an operator from E to itself. By functorial uniqueness,

$$\overline{R_\lambda} = \overline{i_\lambda}\, \widetilde{S_\lambda} : \mathrm{FB}f\mathrm{A}[E] \to \mathrm{FB}f\mathrm{A}[E].$$

Therefore $\overline{R_\lambda} a = 0$ for every λ. By the lemma, $\overline{R_\lambda} a \to a$, and consequently $a = 0$. Thus $\mathrm{FB}f\mathrm{A}[E]$ is semiprime.

The final statement follows from the source theorem that $\mathrm{FB}f\mathrm{A}[E]$ is representable in $C(B_{E^*})$ if and only if it is semiprime [1, Theorem 5.1]. □

Examples Covered

The theorem applies to every Banach space with a monotone Schauder basis, using the canonical initial-segment projections. It also applies to spaces with a monotone finite-dimensional decomposition. More generally, it applies to every Banach space with MAP, because the proof uses finite-rank contractions rather than finite-rank projections. These give positive territory beyond the source's finite-dimensional case and the $L_1(\mu)$ case, although the source question remains open in general.

Verification Notes and Limitations

The proof is formal and non-computational. The key point to review is the strong convergence lemma for induced free algebra maps. It only uses density of $\mathrm{FN}f\mathrm{A}[E]$, the universal property, and norm continuity of the lattice and algebra operations.

The hypothesis is genuinely narrower than the bounded approximation property and the approximation property. Contractivity is used to invoke the universal property of $\mathrm{FB}f\mathrm{A}[E]$ and to make the density argument in the lemma uniform. A uniformly bounded finite-rank approximation with norms bounded by a constant greater than 1 does not directly give homomorphisms on the same free Banach f-algebra, and rescaling the operators destroys the convergence needed at the end. The result therefore does not settle Question 5.10 for general Banach spaces.

Cheap-index searches in the run for arXiv id 2511.13299 and the phrases "free Banach f-algebra", "semiprime", "metric approximation property", "finite-rank contractions", and "representable" found no prior local packet for this strengthened result. Bounded web searches for the source paper and semiprimeness phrases found the source paper and unrelated semiprime Banach algebra material, but no later exact answer.

Human Review Recommendation

Review the functorial factorization through $E_\lambda = R_\lambda(E)$ and the strong-convergence lemma. If both are accepted as consequences of the source universal property and density of $\mathrm{FN}f\mathrm{A}[E]$ in $\mathrm{FB}f\mathrm{A}[E]$, the MAP subcase appears complete. The packet should remain classified as partial unless a separate argument handles Banach spaces without MAP.

References

[1] D. Munoz-Lahoz and P. Tradacete, *Free Banach f-algebras*, arXiv:2511.13299, 2025.